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Project Title:	Sub-regional Assessment of Bone Loss in the Axial Skeleton in Long-Term Spaceflight		
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Zip Code:	94143-0649	**Congressional District:**	8
Comments:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Leblanc, Adrian (Baylor College of Medicine) Genant, Harry (University of California) Lu, Ying (San Francisco)		
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Task Description:	Bone loss is an established risk factor for long-duration space flight. Experience with the Mir missions and long-term bedrest studies has demonstrated the regional extent of this bone loss, its large variability among individuals, and the fact that it persists despite countermeasures which preserve skeletal loading. Despite this important initial progress, several key gaps in our knowledge remain. There is little information about (1) the distribution of this loss between the trabecular and cortical envelopes of the central skeleton, (2) the cortical/trabecular distribution of the recovery of bone mass post-flight, and (3) the extent to which these changes compare to the spread of bone mineral density (BMD) measures in healthy men and women. The goal of this scientific proposal is to address these three key issues. We will achieve these goals by carrying out a longitudinal study of 10-15 individuals making 4-6 month flights on the International Space Station (ISS). We are imaging these subjects pre-flight, immediately postflight, and 1-year postflight using Quantitative Computed Tomography (QCT) of the spine and hip, dual x-ray absorptiometry (DXA) of the spine, hip and calcaneus, and calcaneal ultrasound. We are assessing sub-regional changes in the spine and the femur by analyzing the QCT images using newly-developed algorithms which compute BMD in trabecular and cortical sub-regions in the spine and hip. We will compute bone loss and recovery in these sub-regions and compare the measurements to the corresponding DXA measures which integrate larger volumes of trabecular and cortical bone. This comparison will establish the extent to which DXA measurements underestimate significant local changes which may impact fracture risk. To compare these changes to the variation of bone density in healthy normal populations, we have acquired a reference database consisting of proximal femoral and spinal QCT scans of healthy Caucasian-American males and females between the ages of 35-45, and are in the process of acquiring a comparable reference data set for African-Americans. Finally, in order to establish the utility of calcaneal ultrasound as a surrogate for calcaneal x-ray absorptiometry, we will compare calcaneal ultrasound and DXA measurements. To characterize the compartmental bone changes in biomechanically relevant anatomic sub-regions of the spine and proximal femur associated with long-duration exposure to microgravity and with re-adaptation to earth gravity. To compare these changes with those quantified by DXA images. To characterize the changes in quantitative ultrasound parameters (broadband ultrasound attenuation and speed of sound) of the heel and compare these to changes measured by DXA of the heel. By quantifying the extent of bone loss during the mission and the extent to which it is recovered after the mission, this protocol yields information on the extent to which individuals who have undergone long-duration spaceflight may be subject to long-term increased skeletal frailty. By defining the time course of skeletal recovery, it will yield information on the frequency with which crewmembers may participate in long-duration missions. By examining differences between integral bone measures such as DXA and sub-regional measurements it has impact on the meaning of the DXA information in terms of tracking bone loss.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary goal of our study is to determine the extent to which the skeletal changes occurring in long-duration human spaceflight are concentrated in small sub-volumes of bone or are more generally distributed throughout the bone. If bone loss is concentrated locally, then measures of bone density (such as DXA, the standard clinical measurement method) which integrate across large volumes of bone may not adequately assess the change in bone strength which occurs in long-duration spaceflight and in the recovery period after the flight. The primary benefits of our ground study derive from the acquisition of a normative database of spinal and femoral QCT scans of healthy Caucasian- and African American males and females between the ages of 35-45. First, by providing reference bone mineral density values against which patient measurements may be compared, this experiment will help facilitate the application of QCT of the hip to diagnosis of osteoporosis in the elderly population. With the increasing availability of new CT scanners in outpatient clinics and small hospitals, management of osteoporosis in the growing elderly population may become an important application of these devices. Additionally, the acquisition of reference data for Caucasian-American males and for African-Americans of both genders, will benefit groups which have not been the main focus of osteoporosis research efforts, which have traditionally centered on Caucasian females. A second benefit is that the QCT reference data comparing African- and Caucasian-Americans will provide important data on gender and racial differences in bone density and geometry in the hip and spine. Almost all of the studies in this area have utilized DXA, which measures an areal density depending on both on the mineral content and the size of the bone. Thus, if two racial groups have different bone sizes, DXA may provide incorrect estimates of the racial difference in bone density. By utilizing QCT, on the other hand, we will be able to measure the volumetric bone mineral density in the hip and spine as well as the three-dimensional geometric properties of these bones.
	We have made excellent progress with this research, and the work so far has resulted in two seminal publications in the Journal of Bone and Mineral Research. Aim 1 (paraphrased from the grant proposal): Determine the compartmental distribution of bone loss in the axial skeleton in long-duration spaceflight and determine compare these measures to bone loss rates determined by DXA and quantitative ultrasound. We have completed this aim. We have obtained DXA, QCT and quantitative ultrasound data for pre- and postflight measurements on a total of 18 subjects. QCT measures were performed at the L1 -L2 vertebrae and hip using a volumetric scanning and analysis protocol. DXA was obtained for the L1-L4 spine, the hip and the heel. Quantitative ultrasound was obtained for the heel. Bone loss was established by comparison of pre- and postflight studies. Our initial study, published in the June 2004 issue of the Journal of Bone and Mineral Research measured cortical and trabecular bone loss using QCT of the spine and hip in 14 crewmembers making 4- to 6-month flights on the International Space Station. The crew were composed of study 13 males and 1 female; age range, 40-55 years on long-duration missions (4-6 months) . We used DXA to obtainareal bone mienral density (aBMD) of the hip and spine and volumetric QCT (vQCT) to assess integral, cortical, and trabecular volumetric BMD (vBMD) in the hip and spine. In the heel, DXA was used to measure aBMD, and quantitative ultrasound (QUS) was used to measure speed of sound (SOS) and broadband ultrasound attenuation (BUA). : aBMD was lost at rates of 0.9%/month at the spine ($p < 0.001$) and 1.4-1.5%/month at the hip ($p < 0.001$). Spinal integral vBMD was lost at a rate of 0.9%/month ($p < 0.001$), and trabecular vBMD was lost at 0.7%/month ($p < 0.05$). In contrast to earlier reports, these changes were generalized across the vertebrae and not focused in the posterior elements. In the hip, integral, cortical, and trabecular vBMD was lost at rates of 1.2-1.5%/month ($p < 0.0001$), 0.4-0.5%/month ($p < 0.01$), and 2.2-2.7%/month ($p < 0.001$), respectively. The cortical bone loss in the

Task Progress:	hip occurred primarily by cortical thinning. Calcaneal aBMD measurements by DXA showed smaller mean losses (0.4%/month) than hip or spine measurements, with SOS and BUA showing no change. Our results showed that ISS crewmembers, on average, experienced substantial loss of both trabecular and cortical bone in the hip and somewhat smaller losses in the spine. These results did not support the use of calcaneal aBMD or QUS measurements as surrogate measures to estimate changes in the central skeleton. Aim 2: Recovery of bone following spaceflight We have completed the main part of this aim and are now in the process of competing an extension. The data acquired for this aim have been published this year in the August edition of the Journal of Bone and Mineral Research. This aim involved a total of 16 crewmembers who had complete QCT, DXA and QUS measurements at pre-flight, post-flight and one year after flight. To determine recovery of bone after spaceflight we compared the one year and postflight time points. In these 16 crew members of the International Space Station (ISS) making flights of 4.5-6 months, we used QCT imaging to measure the total, trabecular, and cortical volumetric BMD (vBMD) of the proximal femur. In addition to vBMD, we also quantified BMC, bone volume, femoral neck cross-sectional area (CSA), and femoral neck indices of compressive and bending strength at three time-points: preflight, postflight, and 1 year after mission. We observed that proximal femoral bone mass was substantially recovered in the year after spaceflight, but measures of vBMD and estimated bone strength showed only partial recovery. The recovery of BMC, in the absence of a comparable increase in vBMD, was explained by increases in bone volume and CSA during the year after spaceflight. Our study showed that recovery of bone mass involved an adaptation of the proximal femur to reloading which entailed an increase in bone size and an incomplete recovery of vBMD. The data indicate that recovery of skeletal density after long-duration space missions may exceed 1 year and supports the evidence in the aging literature for periosteal apposition as a compensatory response for bone loss. The extent to which this compensatory effect protects against fracture remains to be seen. In order to shed light on the long-term recovery of bone parameters, 8 subjects have been followed at 3-4 years postflight. The analysis of these data is underway and will be reported in the final report of the study. Aim 3: Normative data We have obtained QCT and DXA images of the spine and hip on 30 young white males, 30 young white females, 25 young black males and 17young black females. The age range of these subjects was 35-45 years. Data on the young white females have been published this year in Bone in a comparison of hip density and geometry (see Meta et al in bibliography) of healthy young females to elderly white females acquired in another study. We observed that the younger subjects had higher bone mass and volumetric BMD in all compartments compared to the elderly subjects, but that the elderly subjects had larger overall bone tissue volume and and femoral neck cross-sectional area, consistent with the idea of periosteal apposition in the femoral neck. We found that cross-sectional age-related changes in the femoral neck differed from those in the trochanter. A comparison of bone geometry and density between gender and race was presented as a poster at the meeting of the American Society of Bone and Mineral Research 2006 in Philadelphia PA. The population standard deviation for trabecular volumetric BMD in the hip was 25-30 gm/cm3 for all four groups. On average, the loss of trabecular BMD was about 17-20 mg/cm3 depending on the hip region, with several subjects having losses exceeding 30 mg/cm3. Conservatively, the mean loss of trabecular vBMD was thus on the order of 0.5-0.6 SD, and our recovery data indicated that this deficit was not substantially recovered. This means that subjects who had a spaceflight, on average, will probably have a T-score half an SD lower than it would have been without their mission. For some subjects, this deficit will be substantially worse.
Bibliography Type:	Description: (Last Updated: 03/20/2017)
Abstracts for Journals and Proceedings	Saeed I, Li J, Lang T. "Ethnic differences in proximal femoral volumetric density and geometry." Presented at the 28th Meeting of the American Society of Bone and Mineral Research, Philadelphia, PA, September 13-17 2006. Journal of Bone and Mineral Research 2006 Suppl:21. , Sep-2006
Articles in Peer-reviewed Journals	Lang TF, Leblanc AD, Evans HJ, Lu Y. "Adaptation of the proximal femur to skeletal reloading after long-duration spaceflight." J Bone Miner Res. 2006 Aug;21(8):1224-30. PMID: 16869720 , Aug-2006
Articles in Peer-reviewed Journals	Meta M, Lu Y, Keyak JH, Lang T. "Young-elderly differences in bone density, geometry and strength indices depend on proximal femur sub-region: a cross sectional study in Caucasian-American women." Bone. 2006 Jul;39(1):152-8. Epub 2006 Feb 3. PMID: 16459156 , Jul-2006